

Network Incident Investigation Report

1. Executive Summary

Field	Value
Investigation Date	2026-07-14
Device(s) Analyzed	Campus-Core-01, Floor3-Access-01, WAN-Router-01
Logs Analyzed	47
Reconstructed Incidents	4 — proven causal chain + RCA
Standalone Alerts	0
Operational Workflows	3
Routine Informational Events	11
Highest Severity Observed	Critical
Overall Investigation Status	Active — Unresolved incidents require immediate attention

4 incident(s) have a proven causal chain and receive full Root Cause Analysis below.

The remaining 14 events are categorized as alerts (0), workflows (3), or routine (11) — none of these are incidents.

2. Investigation Scope

This investigation covers 47 log events collected from 3 device(s) (Campus-Core-01, Floor3-Access-01, WAN-Router-01) between 2026-07-14 09:00:01 UTC and 2026-07-14 09:09:30 UTC. A total of 4 incident(s) were reconstructed with full RCA, alongside 0 standalone alert(s), 3 operational workflow(s), and 11 routine informational event(s) — overall status: Active.

3. Incident Classification Summary

Category	Count	IDs / Labels
Logs Analyzed	47	All parsed log events
Reconstructed Incidents (RCA)	4	INC-0003, INC-0005, INC-0006, INC-0009
Standalone Alerts	0	None
Operational Workflows	3	WORKFLOW-0002, WORKFLOW-0008, WORKFLOW-0013
Routine Informational Events	11	Ntp, Snmp, Dot1X Success, Interface Up, Lldp Neighbor Discovered, Vlan

How to read this table: Only Reconstructed Incidents have a proven causal chain.

Standalone alerts, workflows, and routine events are **not incidents** — classified separately below.

4. Reconstructed Incidents — Full Root Cause Analysis

Only 4 incident(s) below have a proven causal chain.

Each section: Incident Overview (incl. impact) · Timeline · Root Cause · Cause-Effect Chain · Supporting Evidence · Recommendations

Incident INC-0003 — Crc Errors -> Interface Down

4.1.1 Incident Overview

Field	Value
Incident ID	INC-0003
Status	Resolved
Start Time	2026-07-14 09:01:00 UTC
End Time	2026-07-14 09:04:10 UTC
Duration	190.0s
Severity	Warning
Affected Device	Campus-Core-01
Affected Interface(s)	1/1/25
Events in Chain	16
Causal Confidence	75%
Confidence Reasons	✓ Same interface ✓ Recovery observed ✓ Temporal proximity ✓ Known propagation chain

4.1.2 Timeline Reconstruction

2026-07-14 09:01:00 UTC
[WARNING] Crc Errors
Excessive CRC errors detected on interface 1/1/25
|
v

2026-07-14 09:01:04 UTC
[INFO] Interface Down
Link down on interface 1/1/25
|
v

2026-07-14 09:01:05 UTC
[INFO] Interface Down
Interface 1/1/25 operational status changed to DOWN
|
v

```
2026-07-14 09:01:06 UTC
[WARNING] Ospf Neighbor Down
OSPF neighbor 10.1.1.2 on interface 1/1/25 changed state from FULL to DOWN
|
v

2026-07-14 09:01:07 UTC
[INFO] Route Recalculation Started
Routing table recalculation started
|
v

2026-07-14 09:01:08 UTC
[INFO] Route Recalculation Completed
Routing table recalculation completed
|
v

2026-07-14 09:01:10 UTC
[WARNING] Bgp Session Lost
BGP peer 10.1.1.2 session lost
|
v

2026-07-14 09:01:12 UTC
[INFO] Lldp Neighbor Removed
LLDP neighbor removed from port 1/1/25
|
v

2026-07-14 09:01:15 UTC
[WARNING] Stp Topology Change
Instance 0: Topology change detected on port 1/1/25, recalculating spanning tree
```

4.1.3 Recovery Events

Status = **RESOLVED** (Recovery Duration: 190 seconds)

- Interface Up
- Interface Up
- Ospf Neighbor Up
- Bgp Session Established
- Routes Relearned
- Stp Converged
- Lldp Neighbor Discovered

4.1.4 Root Cause Analysis

Field	Detail
Root Cause	Crc Errors
Root Trigger Event	Excessive CRC errors detected on interface 1/1/25

Field	Detail
Device	Campus-Core-01
Causal Confidence	75%
Causal Links Found	19

4.1.5 Propagation

Root Cause: Crc Errors on Campus-Core-01

Propagation:

Crc Errors -> Interface Down -> Interface Down -> Ospf Neighbor Down -> Lldp Neighbor Removed -> Stp Topology Change -> Recovery Completed

4.1.5 Supporting Evidence

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:01:00	4	1/1/25	Warning	Crc Errors	Excessive CRC errors detected on interface 1/1/25
2026-07-14 09:01:04	6	1/1/25	Info	Interface Down	Link down on interface 1/1/25
2026-07-14 09:01:05	7	1/1/25	Info	Interface Down	Interface 1/1/25 operational status changed to DOWN
2026-07-14 09:01:06	8	1/1/25	Warning	Ospf Neighbor Down	OSPF neighbor 10.1.1.2 on interface 1/1/25 changed state from FULL to DOWN
2026-07-14 09:01:07	9	—	Info	Route Recalculation Started	Routing table recalculation started
2026-07-14 09:01:08	10	—	Info	Route Recalculation Completed	Routing table recalculation completed
2026-07-14 09:01:10	11	—	Warning	Bgp Session Lost	BGP peer 10.1.1.2 session lost
2026-07-14 09:01:12	12	1/1/25	Info	Lldp Neighbor Removed	LLDP neighbor removed from port 1/1/25
2026-07-14 09:01:15	13	1/1/25	Warning	Stp Topology Change	Instance 0: Topology change detected on port 1/1/25, recalculating spanning tree
2026-07-14 09:04:00	24	1/1/25	Info	Interface Up	Port 1/1/25 is now on-line

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:04:01	25	1/1/25	Info	Interface Up	Interface 1/1/25 operational status changed to UP
2026-07-14 09:04:02	26	1/1/25	Info	Ospf Neighbor Up	OSPF neighbor 10.1.1.2 on interface 1/1/25 changed state from DOWN to FULL
2026-07-14 09:04:03	27	—	Info	Bgp Session Established	BGP peer 10.1.1.2 session established
2026-07-14 09:04:04	28	—	Info	Routes Relearned	Routes successfully relearned
2026-07-14 09:04:05	29	—	Info	Stp Converged	Instance 0: Topology converged, root bridge unchanged
2026-07-14 09:04:10	30	1/1/25	Info	Lldp Neighbor Discovered	LLDP neighbor discovered on port 1/1/25

4.1.6 Recommendations

#	Action	Rationale
1	Replace cable or transceiver	CRC errors indicate signal integrity problems
2	Check port error counters	Persistent CRC errors may require port replacement

Incident INC-0005 — Dot1X Failure -> Radius Failure

4.2.1 Incident Overview

Field	Value
Incident ID	INC-0005
Status	Active
Start Time	2026-07-14 09:03:00 UTC
End Time	2026-07-14 09:03:20 UTC
Duration	20.0s
Severity	Error
Affected Device	Floor3-Access-01
Affected Interface(s)	1/1/10, 1/1/11
Events in Chain	6
Causal Confidence	0%
Confidence Reasons	✓ Temporal proximity ✓ Known propagation chain

4.2.2 Timeline Reconstruction

2026-07-14 09:03:00 UTC
[ERROR] Dot1X Failure
802.1x: Authentication failed for client AA:BB:CC:DD:EE:01 on port 1/1/10
|
v

2026-07-14 09:03:06 UTC
[ERROR] Dot1X Failure
802.1x: Authentication failed for client AA:BB:CC:DD:EE:02 on port 1/1/11
|
v

2026-07-14 09:03:12 UTC
[ERROR] Radius Failure
RADIUS server 192.168.10.5 unreachable, authentication timeout
|
v

2026-07-14 09:03:15 UTC
[ERROR] Radius Failure
RADIUS server 192.168.10.6 unreachable, authentication timeout
|
v

2026-07-14 09:03:18 UTC
[WARNING] Port Blocked
Port 1/1/10 blocked due to repeated authentication failures
|
v

2026-07-14 09:03:20 UTC
[WARNING] Port Blocked
Port 1/1/11 blocked due to repeated authentication failures

4.2.3 Root Cause Analysis

Field	Detail
Root Cause	Radius Failure
Root Trigger Event	RADIUS server 192.168.10.5 unreachable, authentication timeout
Device	Floor3-Access-01
Causal Confidence	0%
Causal Links Found	12

4.2.5 Supporting Evidence

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:03:00	16	1/1/10	Error	Dot1X Failure	802.1x: Authentication failed for client AA:BB:CC:DD:EE:01 on port 1/1/10

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:03:06	18	1/1/11	Error	Dot1X Failure	802.1x: Authentication failed for client AA:BB:CC:DD:EE:02 on port 1/1/11
2026-07-14 09:03:12	20	—	Error	Radius Failure	RADIUS server 192.168.10.5 unreachable, authentication timeout
2026-07-14 09:03:15	21	—	Error	Radius Failure	RADIUS server 192.168.10.6 unreachable, authentication timeout
2026-07-14 09:03:18	22	1/1/10	Warning	Port Blocked	Port 1/1/10 blocked due to repeated authentication failures
2026-07-14 09:03:20	23	1/1/11	Warning	Port Blocked	Port 1/1/11 blocked due to repeated authentication failures

4.2.6 Recommendations

#	Action	Rationale
1	Investigate root cause device and interface	device=Floor3-Access-01, interface=1/1/10, 1/1/11
2	Add monitoring alerts for this event type	subtype=radius_failure

Incident INC-0006 — Bgp Session Lost -> Route Withdrawal -> Ospf Neighbor Down

4.3.1 Incident Overview

Field	Value
Incident ID	INC-0006
Status	Resolved
Start Time	2026-07-14 09:05:00 UTC
End Time	2026-07-14 09:08:05 UTC
Duration	185.0s
Severity	Warning
Affected Device	WAN-Router-01
Affected Interface(s)	—
Events in Chain	8
Causal Confidence	60%
Confidence Reasons	✓ Recovery observed ✓ Temporal proximity ✓ Known propagation chain

4.3.2 Timeline Reconstruction

2026-07-14 09:05:00 UTC
[WARNING] Bgp Session Lost
BGP peer 203.0.113.1 session lost
|
v

2026-07-14 09:05:02 UTC
[INFO] Route Withdrawal
Route withdrawal initiated for prefix 0.0.0.0/0
|
v

2026-07-14 09:05:05 UTC
[WARNING] Ospf Neighbor Down
OSPF neighbor 10.99.0.1 changed state from FULL to DOWN
|
v

2026-07-14 09:05:08 UTC
[INFO] Route Recalculation Started
Routing table recalculation started
|
v

2026-07-14 09:05:10 UTC
[INFO] Route Recalculation Completed
Routing table recalculation completed

4.3.3 Recovery Events

Status = **RESOLVED** (Recovery Duration: 185 seconds)

- Ospf Neighbor Up
- Bgp Session Established
- Routes Relearned

4.3.4 Root Cause Analysis

Field	Detail
Root Cause	Bgp Session Lost
Root Trigger Event	BGP peer 203.0.113.1 session lost
Device	WAN-Router-01
Causal Confidence	60%
Causal Links Found	5

4.3.5 Propagation

Root Cause: Bgp Session Lost on WAN-Router-01

Propagation:

Route Withdrawal -> Route Recalculation Started -> Route Recalculation Completed -> Recovery Completed

4.3.5 Supporting Evidence

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:05:00	31	—	Warning	Bgp Session Lost	BGP peer 203.0.113.1 session lost
2026-07-14 09:05:02	32	—	Info	Route Withdrawal	Route withdrawal initiated for prefix 0.0.0.0/0
2026-07-14 09:05:05	33	—	Warning	Ospf Neighbor Down	OSPF neighbor 10.99.0.1 changed state from FULL to DOWN
2026-07-14 09:05:08	34	—	Info	Route Recalculation Started	Routing table recalculation started
2026-07-14 09:05:10	35	—	Info	Route Recalculation Completed	Routing table recalculation completed
2026-07-14 09:08:00	43	—	Info	Ospf Neighbor Up	OSPF neighbor 10.99.0.1 changed state from DOWN to FULL
2026-07-14 09:08:02	44	—	Info	Bgp Session Established	BGP peer 203.0.113.1 session established
2026-07-14 09:08:05	45	—	Info	Routes Relearned	Routes successfully relearned

4.3.6 Recommendations

#	Action	Rationale
1	Restore routing adjacency	Verify physical connectivity and protocol timers on WAN-Router-01
2	Check interface stability	Interface — may be causing routing instability

Incident INC-0009 — Ssh Bruteforce -> Ssh Source Blocked

Summary: Repeated SSH authentication failures detected. Automatic source IP blocking was triggered. No successful authentication occurred before mitigation.

4.4.1 Incident Overview

Field	Value
Incident ID	INC-0009

Field	Value
Status	Active
Start Time	2026-07-14 09:07:02 UTC
End Time	2026-07-14 09:07:06 UTC
Duration	4.0s
Severity	Warning
Affected Device	Campus-Core-01
Affected Interface(s)	—
Events in Chain	2
Causal Confidence	60%
Confidence Reasons	✓ Temporal proximity ✓ Known propagation chain

4.4.2 Timeline Reconstruction

```
2026-07-14 09:07:02 UTC
[WARNING] Ssh Bruteforce
SSH login failed from IP 10.99.99.5
|
v

2026-07-14 09:07:06 UTC
[CRITICAL] Ssh Source Blocked
SSH source 10.99.99.5 blocked after maximum failed attempts
```

4.4.3 Root Cause Analysis

Field	Detail
Root Cause	Ssh Bruteforce
Root Trigger Event	SSH login failed from IP 10.99.99.5
Device	Campus-Core-01
Causal Confidence	60%
Causal Links Found	1

4.4.4 Propagation

Root Cause: Ssh Bruteforce on Campus-Core-01

Propagation:

Ssh Bruteforce -> Ssh Source Blocked

4.4.5 Supporting Evidence

Time (UTC)	UID	Interface	Severity	Event Type	Log Message
2026-07-14 09:07:02	40	—	Warning	Ssh Bruteforce	SSH login failed from IP 10.99.99.5
2026-07-14 09:07:06	42	—	Critical	Ssh Source Blocked	SSH source 10.99.99.5 blocked after maximum failed attempts

4.4.6 Recommendations

#	Action	Rationale
1	Block source IP at perimeter firewall	External IP targeting management plane
2	Enable SSH rate limiting	Limit failed login attempts per source
3	Review SSH access control list	Restrict SSH to trusted management IPs only

5. Standalone Alerts

These events have **no proven causal chain**. Each is independent — not part of a cascading incident.

No standalone alerts detected.

6. Operational Workflows

These are **normal, successful operations** — not incidents.

Workflow ID	Event Type	Time (UTC)	Status	Action Required
WORKFLOW-0002	Config Change	2026-07-14 09:00:05	Successful	No
WORKFLOW-0008	Config Change	2026-07-14 09:06:10	Successful	No
WORKFLOW-0008	Config Change	2026-07-14 09:09:20	Successful	No
WORKFLOW-0013	Authentication	2026-07-14 09:09:30	Successful	No

7. Routine Informational Events

These events are **not actionable**. Included for completeness only.

Event Type	Time (UTC)	Device	Notes
Ntp	2026-07-14 09:00:01	Campus-Core-01	Expected periodic time synchronization
Snmp	2026-07-14 09:00:10	Campus-Core-01	Normal SNMP polling session teardown

Event Type	Time (UTC)	Device	Notes
Ntp	2026-07-14 09:02:05	Floor2-Access-01	Expected periodic time synchronization
Dot1X Success	2026-07-14 09:02:00	Floor2-Access-01	Classified as informational by causal engine
Dot1X Success	2026-07-14 09:06:05	Floor2-Access-01	Classified as informational by causal engine
Interface Up	2026-07-14 09:06:00	Floor2-Access-01	Standard port coming online
Lldp Neighbor Discovered	2026-07-14 09:09:10	Floor2-Access-01	Expected neighbor discovery
Snmp	2026-07-14 09:09:00	Campus-Core-01	Normal SNMP polling session teardown
Ntp	2026-07-14 09:09:05	Campus-Core-01	Expected periodic time synchronization
Vlan	2026-07-14 09:09:15	Campus-Core-01	Normal VLAN provisioning activity
Dot1X Success	2026-07-14 09:09:25	Floor3-Access-01	Classified as informational by causal engine

8. Recommendations

Priority	Action	Source
High	Replace cable or transceiver	INC-0003
High	Check port error counters	INC-0003
High	Investigate root cause device and interface	INC-0005
High	Add monitoring alerts for this event type	INC-0005
High	Restore routing adjacency	INC-0006
High	Check interface stability	INC-0006
High	Block source IP at perimeter firewall	INC-0009
High	Enable SSH rate limiting	INC-0009
High	Review SSH access control list	INC-0009
Low	Enrich device vendor metadata for improved classification accuracy	All
Low	Configure alerting for high root-score event subtypes	All

9. Investigation Conclusion

Reconstructed Incidents (4):

- **INC-0003** — Root Cause: Crc Errors | Status: Resolved | Confidence: 75%

- **INC-0005** — Root Cause: Radius Failure | Status: Active | Confidence: N/A
- **INC-0006** — Root Cause: Bgp Session Lost | Status: Resolved | Confidence: 60%
- **INC-0009** — Root Cause: Ssh Bruteforce | Status: Active | Confidence: 60%

Overall Network Health: Poor — active incidents detected

Remaining Uncertainties:

- Causality is inferred from temporal and contextual heuristics — not guaranteed proof.
- Events with vendor = unknown may have reduced classification accuracy.

10. Appendix

Field	Value
Reconstructed Incident IDs	INC-0003, INC-0005, INC-0006, INC-0009
Standalone Alert IDs	None
Total Causal Links	37
Report Generated	2026-07-14T17:53:15Z
Log Reference Files	normalized_events.json, timeline_output.json, causal_inference_output.json
Causality Method	Temporal + contextual heuristics (DAG-graph-partitioned)